MTH2021 Mathematical Methods 2

Assignment 5 – solutions February 16, 2022

1.

$$f(z) = \frac{a_{-m}}{(z-z_0)^m} + \cdots + \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + \cdots$$

Multiply $f(z)$ by $(z-z_0)^m$:

$$(z-z_0)^m f(z) = a_{-m} + \cdots + a_{-1}(z-z_0)^{m-1} + a_0(z-z_0)^m + \cdots$$

Differentiating $(m-1)$ times removes the coefficients a_{-m}, a_{-m+1}, $\cdots$, a_{-2} leaving

$$(m-1)(m-2)\cdots(2)(1)\,a_{-1} + \text{positive powers of } (z-z_0)$$

Taking $z \to z_0$ now removes the terms beyond a_{-1}:

$$\lim_{z\to z_0} \frac{d^{m-1}}{dz^{m-1}}\left[(z-z_0)^m f(z)\right] = (m-1)!\,a_{-1}$$

2. (a) Find the residues of $f(z) = \dfrac{5z-2}{z(z-1)}$ at its poles.

$f(z)$ has simple poles (poles of order 1) at $z = 0$ and $z = 1$. We can use the formula

$$r = \lim_{z\to z_0} \left((z-z_0)\,f(z)\right)$$

At $z_0 = 0$

$$\begin{aligned} r &= \lim_{z\to 0}\left(z\,\frac{(5z-2)}{z(z-1)}\right) \\ &= \lim_{z\to 0}\left(\frac{5z-2}{z-1}\right) \\ &= 2 \end{aligned}$$

At $z_0 = 1$

$$\begin{aligned} r &= \lim_{z\to 1}\left((z-1)\,\frac{(5z-2)}{z(z-1)}\right) \\ &= \lim_{z\to 1}\left(\frac{5z-2}{z}\right) \\ &= 3 \end{aligned}$$

(b) $f(z) = \dfrac{z^2 - 2z + 3}{(z-2)^2}$ has a pole of order 2 at $z = 2$. For the residue we use

$$r = \lim_{z \to z_0} \frac{d}{dz} \left((z - z_0)^2 \, f(z) \right)$$

At $z_0 = 2$

$$\begin{aligned} r &= \lim_{z \to 2} \frac{d}{dz} \left((z-2)^2 \, \frac{(z^2 - 2z + 3)}{(z-2)^2} \right) \\ &= \lim_{z \to 2} \frac{d}{dz} \left(z^2 - 2z + 3 \right) \\ &= \lim_{z \to 2} \, (2z - 2) \\ &= 2 \end{aligned}$$

(c) $f(z) = \dfrac{\sinh z}{z^4}$ has a pole of order 3 at $z = 0$.

To see this, we use the Taylor series for $\sinh z$ about $z = 0$ and divide by z^4 to obtain the Laurent series

$$\begin{aligned} f(z) &= \frac{\sinh z}{z^4} \\ &= \frac{z + \frac{z^3}{3!} + \frac{z^5}{5!} + \cdots}{z^4} \\ &= \frac{1}{z^3} + \frac{1}{3! z} + \frac{z}{5!} + \cdots \end{aligned}$$

Therefore $z = 0$ is a pole of order 3, and the residue (the coefficient of $1/z$) is $1/6$.

3. Show that

(a)

$$\int_0^\infty \frac{x^2}{(a^2 + x^2)^3} \, dx = \frac{\pi}{16 a^3}$$

Since the integrand is even, we can write

$$2 \int_0^\infty \frac{x^2}{(a^2 + x^2)^3} \, dx = \int_{-\infty}^\infty \frac{x^2}{(a^2 + x^2)^3} \, dx$$

The integral on the right is of the form

$$\int_{-\infty}^\infty \frac{P(x)}{Q(x)} \, dx$$

where $P(x)$ and $Q(x)$ are polynomials such that:
• $Q(z)$ has no zeros on the real axis
• the degree of $Q(x)$ is at least 2 greater than that of $P(x)$.
(In this example $P(x) = x^2$, $Q(x) = (a^2 + x^2)^3$.)
From lectures

$$\int_{-\infty}^\infty \frac{P(x)}{Q(x)} \, dx = 2\pi \mathrm{i} \sum (\text{residues of } \frac{P(z)}{Q(z)} \text{ in upper half plane})$$

To find the poles and residues we write

$$f(z) = \frac{P(z)}{Q(z)} = \frac{z^2}{(a^2+z^2)^3} = \frac{z^2}{(-a\mathrm{i}+z)^3(a\mathrm{i}+z)^3}$$

$f(z)$ has poles of order 3 at $z = \pm a\mathrm{i}$. Only the pole at $z = a\mathrm{i}$ is in the upper half plane. The residue at $z = a\mathrm{i}$ is

$$\begin{aligned}
\frac{1}{2!}\lim_{z\to a\mathrm{i}}\frac{d^2}{dz^2}\left((z-a\mathrm{i})^3 f(z)\right) &= \frac{1}{2!}\lim_{z\to a\mathrm{i}}\frac{d^2}{dz^2}\left(\frac{z^2}{(a\mathrm{i}+z)^3}\right)\\
&= \frac{1}{2!}\lim_{z\to a\mathrm{i}}\frac{d}{dz}\left(\frac{2z}{(a\mathrm{i}+z)^3} - \frac{3z^2}{(a\mathrm{i}+z)^4}\right)\\
&= \frac{1}{2!}\lim_{z\to a\mathrm{i}}\frac{d}{dz}\left(\frac{-z^2+2za\mathrm{i}}{(a\mathrm{i}+z)^4}\right)\\
&= \frac{1}{2!}\lim_{z\to a\mathrm{i}}\left(\frac{-2z+2a\mathrm{i}}{(a\mathrm{i}+z)^4} - 4\frac{(-z^2+2za\mathrm{i})}{(a\mathrm{i}+z)^5}\right)\\
&= \frac{1}{2!}\left(-4\frac{(a^2-2a^2)}{(2a\mathrm{i})^5}\right)\\
&= \frac{2a^2}{(2a)^5\mathrm{i}}\\
&= \frac{1}{16a^3\mathrm{i}}
\end{aligned}$$

Therefore

$$\int_{-\infty}^{\infty}\frac{x^2}{(a^2+x^2)^3}\,dx = 2\pi\mathrm{i}\,\frac{1}{16a^3\mathrm{i}} = \frac{\pi}{8a^3}$$

and

$$\int_0^{\infty}\frac{x^2}{(a^2+x^2)^3}\,dx = \frac{\pi}{16a^3}$$

An alternative way to get the residue is directly from the Laurent expansion about the pole at $z = a\mathrm{i}$. We seek an expansion in powers of $w = z - a\mathrm{i}$:

$$\begin{aligned}
f(z) &= \frac{z^2}{(-a\mathrm{i}+z)^3(a\mathrm{i}+z)^3}\\
&= \frac{(w+a\mathrm{i})^2}{w^3(2a\mathrm{i}+w)^3}\\
&= \frac{(w^2+2a\mathrm{i}w-a^2)}{w^3}\,\frac{1}{(2a\mathrm{i}+w)^3}\\
&= \frac{1}{(2a\mathrm{i})^3}\left(\frac{1}{w}+\frac{2a\mathrm{i}}{w^2}-\frac{a^2}{w^3}\right)\left(1+\frac{w}{2a\mathrm{i}}\right)^{-3}\\
&= \frac{1}{(2a\mathrm{i})^3}\left(\frac{1}{w}+\frac{2a\mathrm{i}}{w^2}-\frac{a^2}{w^3}\right)\left(1-3\frac{w}{2a\mathrm{i}}+\frac{3\times 4}{2}\left(\frac{w}{2a\mathrm{i}}\right)^2-\cdots\right)
\end{aligned}$$

using the binomial expansion which converges for z near the pole. This is sufficient to determine the residue which is the coefficient of $\frac{1}{w}$. Collecting the terms in $\frac{1}{w}$ gives for the residue

$$\frac{1}{(2a\mathrm{i})^3}\left(1-3+\tfrac{3}{2}\right) = \frac{1}{16a^3\mathrm{i}}$$

(b) The integral

$$\int_{-\infty}^{\infty} \frac{x^2}{1+x^4}\, dx$$

is of the form

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)}\, dx$$

where $P(x)$ and $Q(x)$ are polynomials such that:
- $Q(z)$ has no zeros on the real axis
- the degree of $Q(x)$ is at least 2 greater than that of $P(x)$.

(In this example $P(x) = x^2$, $Q(x) = 1 + x^4$.)
From lectures

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)}\, dx = 2\pi\mathrm{i} \sum \left(\text{residues of } \frac{P(z)}{Q(z)} \text{ in upper half plane}\right)$$

The poles of the integrand occur whenever $z^4 + 1 = 0$, that is, at the four fourth roots of -1:

$$z_1 = \exp(\mathrm{i}\pi/4) = \frac{1+\mathrm{i}}{\sqrt{2}} \quad , \quad z_2 = \exp(3\pi\mathrm{i}/4) = \frac{-1+\mathrm{i}}{\sqrt{2}}$$
$$z_3 = \exp(5\pi\mathrm{i}/4) = \frac{-1-\mathrm{i}}{\sqrt{2}} \quad , \quad z_4 = \exp(7\pi\mathrm{i}/4) = \frac{1-\mathrm{i}}{\sqrt{2}}$$

Of these poles, only two – those at z_1 and z_2 – lie in the upper half plane. Both are simple poles (poles of order 1). To find the residues we can write

$$f(z) = \frac{P(z)}{Q(z)} = \frac{z^2}{1+z^4} = \frac{z^2}{(z-z_1)(z-z_2)(z-z_3)(z-z_4)}$$

The residue at z_1 is

$$\begin{aligned} r_1 &= \lim_{z\to z_1} \left((z-z_1)\, f(z)\right) \\ &= \lim_{z\to z_1} \left(\frac{z^2}{(z-z_2)(z-z_3)(z-z_4)}\right) \\ &= \frac{z_1^2}{(z_1-z_2)(z_1-z_3)(z_1-z_4)} = \frac{1}{2\sqrt{2}(1+\mathrm{i})} \end{aligned}$$

The residue at z_2 is

$$\begin{aligned} r_2 &= \lim_{z\to z_2} \left((z-z_2)\, f(z)\right) \\ &= \lim_{z\to z_2} \left(\frac{z^2}{(z-z_1)(z-z_3)(z-z_4)}\right) \\ &= \frac{z_2^2}{(z_2-z_1)(z_2-z_3)(z_2-z_4)} = \frac{1}{2\sqrt{2}(-1+\mathrm{i})} \end{aligned}$$

Thus

$$\int_{-\infty}^{\infty} \frac{x^2}{1+x^4}\, dx = 2\pi\mathrm{i}(r_1 + r_2) = \frac{\pi}{\sqrt{2}}$$

4. If $z = \exp(\mathrm{i}\theta)$, then $dz = \mathrm{i}z\, d\theta$ and $\cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right)$. Hence

$$\begin{aligned}
I &= \int_0^{2\pi} \frac{d\theta}{5 + 4\cos\theta} \\
&= \oint_C \frac{dz}{\mathrm{i}z} \frac{1}{5 + 2\left(z + \frac{1}{z}\right)} \\
&= \frac{1}{\mathrm{i}} \oint_C \frac{1}{5z + 2z^2 + 2}\, dz \\
&= \frac{1}{\mathrm{i}} \oint_C \frac{1}{(2z+1)(z+2)}\, dz
\end{aligned}$$

The contour C is the unit circle $|z| = 1$.

The integrand has poles of order 1 at $z = -\frac{1}{2}$ and $z = -2$. Only $z = -\frac{1}{2}$ lies within C. The residue at this pole is

$$\lim_{z \to -\frac{1}{2}} \left((z + 1/2) \frac{1}{(2z+1)(z+2)} \right) = \lim_{z \to -\frac{1}{2}} \left(\frac{1}{2(z+2)} \right) = \frac{1}{2(-\frac{1}{2} + 2)} = \frac{1}{3}$$

Therefore by the residue theorem

$$\begin{aligned}
I &= \frac{1}{\mathrm{i}} \oint_C \frac{1}{(2z+1)(z+2)}\, dz \\
&= \frac{1}{\mathrm{i}}\, 2\pi\mathrm{i}\, \frac{1}{3} \\
&= \frac{2\pi}{3}
\end{aligned}$$

5. Consider the integral

$$I = \int_{-\infty}^{\infty} \frac{\exp(\mathrm{i}x)}{(1+x^2)^2}\, dx$$

It is of the form

$$I = \int_{-\infty}^{\infty} \exp(\mathrm{i}\lambda x) \frac{P(x)}{Q(x)}\, dx \qquad (1)$$

where $P(x)$ and $Q(x)$ are polynomials such that:
• $Q(z)$ has no zeros on the real axis
• the degree of $Q(x)$ is at least 1 greater than the degree of $P(x)$
• $\lambda > 0$.
(In this example $P(x) = 1$, $Q(x) = (1+x^2)^2$, $\lambda = 1$.)

From lectures

$$\int_{-\infty}^{\infty} \exp(\mathrm{i}\lambda x) \frac{P(x)}{Q(x)}\, dx = 2\pi\mathrm{i} \sum \left(\text{residues of } \exp(\mathrm{i}\lambda z) \frac{P(z)}{Q(z)} \text{ in upper half plane}\right)$$

In our case

$$\exp(\mathrm{i}\lambda z)\,\frac{P(z)}{Q(z)} = \frac{\exp(\mathrm{i}z)}{(1+z^2)^2} \quad (2)$$

$$= \frac{\exp(\mathrm{i}z)}{(z-\mathrm{i})^2(z+\mathrm{i})^2} \quad (3)$$

has poles of order 2 at $z = \pm\mathrm{i}$. Only one – at i – is in the upper half plane. The residue at $z = \mathrm{i}$ is

$$r = \frac{1}{1!}\lim_{z\to\mathrm{i}}\frac{d}{dz}\left[\frac{(z-\mathrm{i})^2\exp(\mathrm{i}z)}{(z-\mathrm{i})^2(z+\mathrm{i})^2}\right] = -\frac{\mathrm{i}\exp(-1)}{2} \quad (4)$$

Therefore

$$\int_{-\infty}^{\infty}\frac{\exp(\mathrm{i}x)}{(1+x^2)^2}\,dx = 2\pi\mathrm{i}r = \pi\ \exp(-1) \quad (5)$$

Taking real parts:

$$\int_{-\infty}^{\infty}\frac{\cos x}{(1+x^2)^2}\,dx = \frac{\pi}{\mathrm{e}} \quad (6)$$